

What Does an SVM Learn from Blogs Written by People with Alzheimer's?

Experimental Report

Chiara Tosadori
Machine Learning for NLP
University Of Trento

August 17, 2026

1. Introduction

Text classification systems can achieve very high performance even when part of their predictive power derives from dataset features that are only indirectly related to the phenomenon of interest. This can also happen in the 5th practical : SVMs for Alzheimer's detection, due to the structure of its dataset. The dataset consists, in fact, of blog posts divided into two classes: posts associated with bloggers with Alzheimer's and posts associated with relatives, caregivers, or unaffected controls. Because of this, the classifier's excellent performance could be attributed to shortcuts related to discourse role, different writing perspectives (patient vs. caregiver), the writing style of a specific blogger, and the lexical topic of the posts, rather than to the recognition of cognitive decline.

From this observation arose the question that guided this experiment:

Does the classifier detect Alzheimer related linguistic patterns, or does it mainly exploit dataset-specific shortcuts?

However, due to the very constitution of the dataset (lack of author information, etc.), the experiment was not able to fully answer the question, but rather focused on discovering how much the model relied on the structural elements of his text, rather than on its specific lexical characteristics. Thus, with regard to Alzheimer's-specific markers, any interpretation of these results remains speculative.

To do so, three different ways of representing the same texts were compared, always using the same classifier (linear SVM). The first representation used a standard TF-IDF, preserving the entire lexical content of each post. The second was a delexicalized representation, in which content words were replaced by their grammatical category (POS), while still maintaining function words, pronouns, and punctuation. The third represented each post through a reduced set of interpretable and aggregated linguistic features, chosen based on the psycholinguistic literature on Alzheimer.

The full TF-IDF representation is expected to achieve the best performance, as it can exploit both corpus-specific lexical features and the linguistic information extractable from it. However, if the model is still able to classify well even with a much smaller number of features, this could mean that the differences between the two classes are not limited to individual words and the topics covered, and that these differences are themselves discriminative information.

2. Data and Model

The dataset consists of two classes. Class 1 contains posts from bloggers affected by Alzheimer, while Class 2 (labeled Class 0) consists of posts written by the control group.

Seven empty documents were removed from the original practical dataset, resulting in a dataset of 3,644 posts, of which 2,274 belong to Class 1 and 1,370 to Class 2.

All experiments used linear SVM. This choice is motivated by the fact that the project's goal is not to achieve the highest possible performance but rather to compare different representations of the same documents using the same classification method in an easy and understandable way.

3. Experimental Setup

The project was developed in three phases:

1. Data preparation and understanding	2. Baseline construction	3. Experiment
• Analysis of the number of documents • Verification of empty documents • Verification of lost or duplicate documents • Analysis of obvious differences between the two classes	• Division into training and test sets • TF-IDF on the training set only • Training the SVM • 5-fold cross-validation • Generation of predictions	• Removal of clinical words • Removal of blog/web artifacts • Removal of narrative/discursive role • Delexicalization • Interpretable linguistic features

3.1 Baseline

The original practical SVM was used as the initial analysis point, but its evaluation pipeline was modified before running the main experiments.

Before the train/test split, the original code applied some preprocessing steps and used a non-standard TF-IDF representation computed at the class level rather than per document. Evaluation also relied on a single random split. These choices made it difficult to understand how well the model truly generalizes and to fairly compare the different representations.

For this reason, a new baseline was implemented using a standard TF-IDF applied at the individual document level, followed by a linear SVM. To avoid information leakage between training and testing, the TF-IDF was configured solely on the training data of each fold.

It is important to underline that it would have been better to perform a split at the author level rather than at the document level, because by doing so there would not have been texts by the same author in both the training and the test sets, however this was not possible because the authors were not provided with the database.

All representations were evaluated using the same 5-fold stratified cross-validation, with shuffling and a fixed random seed of 42. The SVM uses $C = 1.0$ and balanced class weights. In addition to accuracy, balanced accuracy and macro-F1 (chosen as the primary metric) were used as evaluation metrics. The latter two are particularly suitable in contexts like this, characterized by unbalanced classes.

3.2 TF-IDF Representation and Ablation Analysis

The first model was the baseline and therefore represented each post using TF-IDF features. This representation maintained the full lexical content of the posts and achieved a Macro-F1 score of 0.98. This result is, however, suspiciously high for a classifier that is supposed to detect posts written by people with Alzheimer’s, raising concerns about the use of potential shortcuts, thus motivating the desire to investigate which factors are driving the model’s performance.

Therefore, to carry out a brief preliminary investigation, the SVM coefficients were analyzed. This analysis showed that many of the most discriminative features distinguishing the two classes were not directly linked to linguistic differences associated with Alzheimer’s disease.

Based on this result, three experimental ablations were conducted (see Appendix A for further details). A substantial drop in performance following these ablations would have suggested that the model was relying primarily on a small number of simple shortcuts. However, performance decreased only marginally. This suggests either that the predictive signal is distributed across the vocabulary rather than concentrated in a few confounding terms, or that the model relies on other types of information that are not captured by these ablations.

3.3 Delexicalized Representation

To test more directly how much the model depended on the lexicon, a second representation was created with the delexicalization of the posts. If the model’s performance remains high even after delexicalization, this could suggest that the classification is based on structural and syntactic patterns rather than specific lexical content. Conversely, if delexicalization caused a significant decrease of the performance, it would mean that the information signal was predominantly semantic and distributed across the vocabulary.

Delexicalization removes explicit semantic content but does not alter the idiosyncratic writing style, which is instead isolated and amplified. As a result, higher performance after delexicalization only tells us that the model is relying on structural information; it is not a proof that it is capturing signs of cognitive decline since the maintenance of high performance is compatible both with genuine clinical markers of Alzheimer and with the model learning individual bloggers’ writing styles.

To test this representation, spaCy’s part-of-speech tagging was used. Content words were replaced with their POS tags, while function words and punctuation were retained unchanged. Then, the same TF-IDF + Linear SVM architecture was applied to these delexicalized sequences.

3.4 Linguistic only model

The third representation compressed each post into 22 linguistic features. These features were selected a priori, on a theoretical basis, to cover two main aspects: different linguistic dimensions and typical signs of cognitive decline in writing. Only afterwards was their presence in the raw data and their individual predictive value evaluated (for further details, see Appendix B).

The features were organized into five groups:

- **Lexical features:** Average word length, Moving-average type-token ratio (MATTR), Repetition rate and Vague-word rate.
- **Syntactic features:** Average sentence length, Sentence-length variability, Average dependency depth, Subordination rate and Clause-to-sentence ratio.
- **POS/reference:** Pronoun rate, Noun/pronoun ratio, Noun rate, Verb rate, Adjective rate and Adverb rate and Function-word rate.
- **Discourse features:** Lexical overlap between adjacent sentences, Connective rate and Hedge rate.
- **Tension/modal:** Modal-Verb Rate, Past-Tense Rate and Present-Tense Rate.

Before classification, all 22 features were standardized using statistics computed only from the training fold. Each post was then represented through these features and then classified through the linear SVM and cross-validation protocol used for the other representations.

4. Results

4.1 Comparison Across Representations

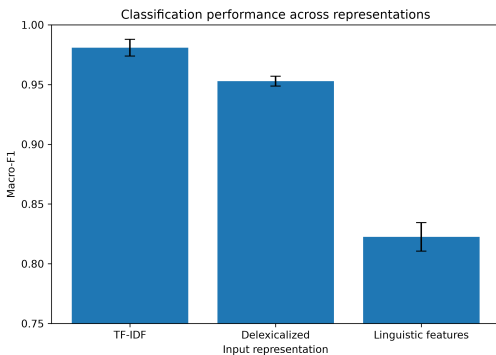


Table 1: Classification results across representations.

Representation	Macro-F1	Std.	Bal. Acc.
TF-IDF	0.9808	0.0070	0.9821
Delexicalized	0.9529	0.0041	0.9588
Linguistic features	0.8225	0.0119	0.8325

The best performance was achieved by the TF-IDF representation, using approximately 26,000 lexical features, with a Macro-F1 score of 0.9808.

Delexicalization produced a score of 0.9529 (a drop of 2.8 percentage points). This decrease is meaningful, as it exceeds the standard deviation; nonetheless, it does not undermine the model’s ability to still achieve near-perfect classification.

The linguistic-only model, which compressed each individual post into just 22 numbers, achieved a Macro F1 score of 0.8225, suggesting that a substantial portion of the separation between the groups can be captured with aggregated and interpretable linguistic properties.

While confirming the presence of a significant linguistic signal, these aggregated features could not only represent cognitive differences but simultaneously capture multiple confounding factors such as individual style, patient/caregiver roles, and differences between blogs. Therefore, it was deemed relevant to verify which of the 22 features was the most discriminating.

4.2 Linguistic Feature Groups

The 22 features were first divided into their five thematic families, and separate models were then trained using only one family at a time.

Table 2: Performance of the SVM using individual linguistic feature groups.

Feature group	N. features	Macro-F1	Std.	Bal. Acc.
All 22	22	0.8225	0.0119	0.8325
POS/reference	7	0.7299	0.0121	0.7351
Lexical	4	0.6960	0.0074	0.6992
Syntactic	5	0.6811	0.0143	0.6859
Discourse	3	0.6299	0.0212	0.6389
Tense/modal	3	0.6042	0.0167	0.6100

Not a single thematic group came close to matching the performance of the full set. In fact, combining the groups adds approximately 9 macro-F1 points compared to the best single group. This may mean that the model is not relying on one feature or shortcut, but is getting information from all these linguistic dimensions, dimensions that likely contain complementary but not redundant information.

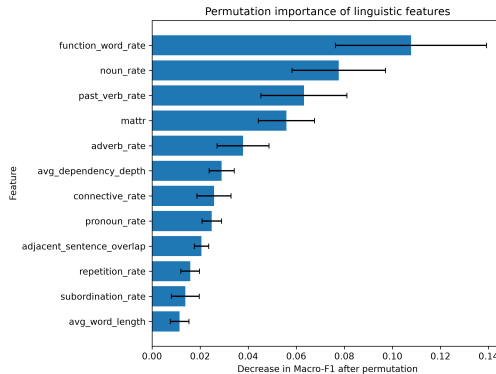
The POS/reference family was significantly stronger on its own, which could mean that it contains the most discriminative signal. However, the signal should be interpreted with caution, as this family also contains the most features, so a direct comparison is not entirely fair.

The weakest families are still scored above 0.60, indicating that they are not useless, especially for the complete model.

4.3 Individual Feature Analysis

After this, the focus shifted to individual features. To do so, we performed 5-fold cross-validation and extracted the SVM coefficients for each fold, computing their mean and standard deviation. Some features were found to be stable in sign, with small deviations, and associated consistently with either the Alzheimer or the control group. Other features, however, proved less robust within the classifier, showing a small mean coefficient relative to their variability and even changing sign across folds. (see Appendix C)

Some features, however, are highly complementary, so it is possible that the SVM distributes weight among them in ways that are not intuitive. To address these problems, two additional analyses were performed: correlation analysis, using Pearson’s correlation to identify features measuring similar information, and permutation importance, to assess how much performance actually depends on each individual feature.



Thanks to the permutation it was discovered that the model appears to exploit four complementary types of information: grammatical composition, lexical variety, syntactic complexity, and discourse structure.

It was also possible to identify the least consistent features: modal-verb rate, clause-to-sentence ratio, hedge rate, noun/pronoun ratio, average sentence length, and sentence-length standard deviation.

One of these less relevant features was average sentence length. Its case is very interesting because when analyzing the dataset, length emerged as a discriminant between the two classes. However, when compared with

other syntactic measures such as dependency depth and subordination rate, it disappears. This effect is very interesting because it seems to suggest that it is not length but rather the underlying syntactic structure that provides information.

The results of the correlation analysis (see Appendix D) also confirmed the presence of correlations among some features, whose coefficients and permutations therefore cannot be interpreted as strictly independent.

4.4 Error Analysis

An error analysis was conducted on the linguistic-only model using out-of-fold predictions to understand which posts the model misclassifies and why. It emerged that many high-confidence errors correspond to very short texts on which the 22 features are unstable. Furthermore, some of these documents were not personal posts but rather titles, announcements, emails, or automated blog content (details in Appendix E).

Text length was therefore found to be a significant confounder for linguistic features. It was therefore considered appropriate to investigate this phenomenon further, by checking how much the model’s performance worsened on short posts (Appendix F), and how it performed when restricted to long posts only (Appendix G). The results showed that performance improves progressively as shorter documents are excluded. The main result, however, remained the one obtained on the full dataset, because restricting the analysis to texts longer than 500 words would leave the control class with roughly three times as many documents as the Alzheimer class, substantially changing the distribution between classes.

A final check, based on the comparison of errors between models (Appendix H), showed that:

- Within the Alzheimer class there are articles, press releases, emails, information manuals, etc., so not all the tests on the Alzheimer blog are written by people with Alzheimer;
- In some cases, the structural signals are sufficient to correct an incorrect lexical decision.

5. Discussion

The experiment showed that the chosen representation radically changes what the classifier is able to exploit.

The full TF-IDF representation achieved near-perfect performance. Analysis of its coefficients showed that part of its effectiveness is attributable to lexical information not directly related to Alzheimer’s disease. However, removing small groups of these terms did not lead to a substantial drop in performance, suggesting that the lexical signal is redundant and distributed across a large portion of the vocabulary.

Delexicalization strengthened this interpretation. Despite the removal of content words, the model performance remained very high. This indicates that individual content words are not necessary for good classification, but grammatical function words, punctuation, and POS sequences alone are sufficient to distinguish the two classes. However, this signal is also ambiguous: this type of result can be obtained whether the classifier has identified a genuine cognitive effect, individual writing style, or discourse role.

Finally, the linguistic-only model achieved the lowest performance of the three, a result that was expected given the significantly smaller amount of information available. However, its result remained very good, demonstrating that this properties discriminate the classes well. The fact that no single feature group matches the performance of the full 22-feature model further suggests that the signal is distributed across several complementary dimensions.

Two important limitations emerged from the error analysis:

- Features become unstable on very short texts;
- The Alzheimer class included not only posts written by people with cognitive decline, but also articles, emails, or informational material. This means that the Alzheimer class is partially contaminated by non-representative content.

Overall, the results obtained suggest that the dataset contains several layers of predictive information. However, for the reasons above, none of these results should be interpreted as clinical markers of Alzheimer’s disease. This is because this paper is a classification study and it is unable to separate cognitive effects from corpus-specific ones such as writing style or narrative role. It would therefore be interesting to further investigate these results with a cross-validation split on authors to see whether the performance holds on bloggers never seen in training.

6. Appendix

6.1 Appendix A: Lexical Ablations

The 50 most relevant coefficients were extracted for both the Alzheimer class and the control class. Among these, some terms were identified because they could be predictive without necessarily being linked to linguistic decline. These features were subsequently grouped into three possible macro-categories of shortcuts:

- **Identity/web domain:** terms related to personal identity and references to the blogosphere;
- **Discourse Role:** pronouns and kinship terms;
- **Clinical domain:** lexicon related to medical and diagnostic aspects.

Table 3: Terms removed in the three lexical ablation experiments.

Identity / Web	Discourse Role	Clinical Topic
joe, alan, vince, jim, fisher, rodney, atlanta, georgia, athens, missouri, sedalia, blogspot, copyright, http, com, org, facebook, helpparentsagewell	mom, mother, dad, father, parents, parent, wife, husband, son, daughter, kids, child, children, she, her, hers, he, him, his, me, my, mine, ours	alzheimer, alzheimers, Alzheimer, ad, lbd, lewy, earlyonset, onset, aging, disease, diagnosis, diagnosed, patient, patients, caregiver, caregivers, caregiving, memory, cognitive, neurologist, neurologists, namenda, aricept, razadyne, medicare, elders

With these categories three experimental ablations were conducted, one for each. It is important to note that the selection of these vocabularies was based on a manual, curated ablation analysis: the model was first trained to identify the coefficients with the highest absolute values, and only subsequently were the related hypotheses formulated.

The results are reported in Table 4.

Table 4: Macro-F1 results for the lexical ablation experiments.

Experiment	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean	Std.
Baseline	0.9781	0.9811	0.9927	0.9810	0.9709	0.9808	0.0070
Without blog/web artifacts	0.9766	0.9782	0.9898	0.9796	0.9693	0.9787	0.0066
Without discourse-role cues	0.9796	0.9738	0.9912	0.9751	0.9695	0.9779	0.0074
Without clinical topic words	0.9781	0.9782	0.9927	0.9825	0.9709	0.9805	0.0072

In the removal of identity and web-related artifacts, a decline in performance was observed in all five folds, which, however, remained below the model’s standard deviation. This stability could be due to the presence of redundant signals in the vocabulary (other features capable of capturing a similar or related information signal).

The second ablation phase, focused on removing discourse role cues, also showed a reduced effect. Although the impact was slightly greater than that one observed for blog features, the decline still remained below the standard deviation. Unlike the first case, however, performance was not uniform across the various folds: some showed a slight improvement, while others showed a deterioration.

Finally, removing the clinical lexicon did not produce substantial changes in performance; all the 5 fold performed as good as the baseline, if not slightly better.

The fact that the model did not significantly lose accuracy following the systematic removal of these three groups of high-importance features suggests that the predictive signal is distributed across the entire vocabulary.

6.2 Appendix B: Linguistic Features

The 22 linguistic features were compared across the two classes before classification.

Table 5: Descriptive statistics of the 22 linguistic features for the Alzheimer and control classes.

Feature	Alzheimer		Control	
	Mean	Med.	Mean	Med.
Avg. sentence length	14.1973	13.8545	16.4469	16.0873
Sentence length std.	7.4176	6.9986	8.5056	7.9071
Avg. dependency depth	4.2296	4.1847	4.7836	4.7439
Subordination rate	1.1616	1.1017	1.3961	1.3422
Clause/sentence ratio	1.7831	1.7778	1.8736	1.9070
Avg. word length	4.1065	4.0258	4.3692	4.3296
MATTR	0.7683	0.7595	0.7878	0.7811
Repetition rate	0.1339	0.1350	0.1428	0.1429
Pronoun rate	0.1611	0.1667	0.1311	0.1310
Noun/pronoun ratio	1.6405	1.2031	2.3531	1.8958
Noun rate	0.2269	0.2046	0.2521	0.2457

Feature	Alzheimer		Control	
	Mean	Med.	Mean	Med.
Verb rate	0.1396	0.1429	0.1485	0.1477
Adjective rate	0.0594	0.0593	0.0726	0.0674
Adverb rate	0.0597	0.0598	0.0491	0.0485
Function-word rate	0.5067	0.5190	0.4666	0.4755
Vague-word rate	0.0039	0.0016	0.0027	0.0015
Adjacent sent. overlap	0.0292	0.0271	0.0378	0.0365
Connective rate	0.2886	0.2857	0.2282	0.2124
Hedge rate	0.0045	0.0028	0.0036	0.0026
Modal-verb rate	0.0149	0.0139	0.0141	0.0133
Past-verb rate	0.3523	0.3333	0.4563	0.4624
Present-verb rate	0.6288	0.6522	0.5364	0.5333

The Alzheimer class appears to have shorter sentences, slightly shorter words, more pronouns, a lower pronoun-to-noun ratio, a lower MATTR, slightly more frequent vague words, less variability in sentence length, slightly shallower syntactic trees, slightly fewer subordinate clauses, fewer adjectives, less sharing of content words in consecutive sentences on average, more function words, more verbs in the present tense and fewer verbs in the past, a higher hedge rate, and a slightly lower clause-to-sentence ratio.

All these syntactic features behave consistently with Alzheimer’s literature, which shows that the writing of Alzheimer’s patients has an average simpler structure. Another positive detail is that the mean and median are quite close for almost all features, suggesting that this difference is not caused by some extreme document (except for the noun-to-pronoun ratio).

There are, however, a few features that do not behave as one might expect under the Alzheimer’s hypothesis:

- **Repetition rate.** Repetition is higher in the control groups. However, this feature is complemented by MATTR, which is lower for Alzheimer’s. This may mean that a text with low overall diversity can occur without specific content words being frequently repeated.
- **Connective rate.** The Alzheimer class has more connectives. This too, however, has a possible explanation: among the list of the most significant coefficients for the Alzheimer class there were “But” and “So”, suggesting that people with Alzheimer may use simple, recurrent connectives more frequently. This result would therefore still be consistent with the Alzheimer’s hypothesis, especially when combined, as in this case, with shorter sentences, less subordination, and lower dependency depth.

Although these features behave consistently with the Alzheimer’s literature, it is important to remember that they do not identify the condition with certainty, some of these patterns are compatible with alternative interpretations: the frequency of nouns, adverbs, and adjectives may depend on individual writing style, just as the more frequent use of the present tense among people with Alzheimer’s could stem from their role as patients, who tend to talk more about their own everyday personal experience as opposed to the control group, which also includes caregivers, who tend to recount other people’s experiences in the third person and in the past tense.

NB: The features related to the cognitive dimension were inspired by “A systematic literature review of automatic Alzheimer’s disease detection from speech and language” (Petti, Baker, Korhonen, 2020) and “Artificial intelligence-driven natural language processing for identifying linguistic patterns in Alzheimer’s disease and mild cognitive impairment: A study of lexical, syntactic, and cohesive features of speech through picture description tasks” (Nyongesa, Hogarth, Pa, 2025).

6.3 Appendix C: Feature Importance

The stability and direction of the 22 linguistic features were analyzed by extracting the SVM coefficients independently for each cross-validation fold.

Table 6: Mean SVM coefficients and variability across the five folds.

Feature	Mean coef.	Std.	Direction
noun_rate	0.6998	0.1187	Alzheimer
function_word_rate	0.4653	0.1210	Alzheimer
matr	-0.3243	0.0257	Control
past_verb_rate	-0.2891	0.0455	Control
adverb_rate	0.2268	0.0470	Alzheimer
avg_dependency_depth	-0.1902	0.0151	Control
pronoun_rate	0.1755	0.0119	Alzheimer
connective_rate	0.1657	0.0219	Alzheimer
adjacent_sentence_overlap	-0.1632	0.0146	Control
repetition_rate	-0.1556	0.0240	Control
subordination_rate	-0.1238	0.0324	Control
avg_word_length	-0.1083	0.0197	Control
vague_word_rate	0.0867	0.0098	Alzheimer
adjective_rate	-0.0815	0.0414	Control
present_verb_rate	0.0540	0.0499	Alzheimer
noun_pronoun_ratio	0.0419	0.0234	Alzheimer
sentence_length_std	-0.0349	0.0100	Control
verb_rate	-0.0284	0.0598	Control
avg_sentence_length	-0.0183	0.0310	Control
clause_to_sentence_ratio	-0.0148	0.0350	Control
hedge_rate	-0.0116	0.0081	Control
modal_verb_rate	0.0029	0.0173	Alzheimer

Table 7 reports the coefficient obtained for each individual fold.

Table 7: SVM coefficients across the five cross-validation folds.

Feature	F1	F2	F3	F4	F5
noun_rate	.4949	.7816	.8436	.6674	.7113
function_word_rate	.2704	.5445	.6225	.4036	.4854
matr	-.3688	-.3074	-.3378	-.3059	-.3014
past_verb_rate	-.2842	-.2445	-.2971	-.2489	-.3707
adverb_rate	.1425	.2461	.2752	.2118	.2582
avg_dependency_depth	-.2030	-.1884	-.2114	-.1756	-.1725
pronoun_rate	.1859	.1594	.1862	.1834	.1628
connective_rate	.1923	.1522	.1541	.1388	.1911
adjacent_sentence_overlap	-.1432	-.1540	-.1752	-.1838	-.1599
repetition_rate	-.1999	-.1584	-.1315	-.1383	-.1498
subordination_rate	-.1395	-.1465	-.0692	-.1582	-.1057

Feature	F1	F2	F3	F4	F5
avg_word_length	-.1197	-.0842	-.0990	-.1405	-.0978
vague_word_rate	.0937	.0856	.0722	.0815	.1006
adjective_rate	-.1545	-.0765	-.0261	-.0813	-.0688
present_verb_rate	.0683	.1112	.0355	.0876	.0327
noun_pronoun_ratio	.0329	.0233	.0874	.0270	.0389
sentence_length_std	-.0310	-.0253	-.0430	-.0502	-.0249
verb_rate	-.1405	-.0003	.0374	-.0194	-.0190
avg_sentence_length	-.0195	-.0221	.0149	.0082	-.0730
clause_to_sentence_ratio	.0091	.0121	.0832	-.0066	-.0055
hedge_rate	-.0018	-.0075	-.0063	-.0217	-.0209
modal_verb_rate	.0327	-.0151	-.0088	.0115	-.0060

6.4 Appendix D: Correlation and Permutation Importance

Pearson’s correlation was computed between the 22 linguistic features to identify variables capturing overlapping information. Table 8 reports the highest correlations. Strong correlations ($|r| \geq 0.70$) are highlighted in bold.

Table 8: Highest Pearson correlations between linguistic features.

Feature 1	Feature 2	r
past_verb_rate	present_verb_rate	-0.8324
noun_rate	function_word_rate	-0.8279
avg_sentence_length	avg_dependency_depth	0.8028
subordination_rate	clause_to_sentence_ratio	0.7805
avg_sentence_length	subordination_rate	0.7756
pronoun_rate	function_word_rate	0.7585
avg_word_length	function_word_rate	-0.7411
pronoun_rate	noun_rate	-0.7095
avg_word_length	noun_rate	0.7034
avg_word_length	pronoun_rate	-0.6963
avg_sentence_length	clause_to_sentence_ratio	0.6737
matr	repetition_rate	-0.6416
avg_sentence_length	sentence_length_std	0.6403
repetition_rate	adjacent_sentence_overlap	0.6352
pronoun_rate	noun_pronoun_ratio	-0.6151
matr	function_word_rate	-0.6151
avg_dependency_depth	subordination_rate	0.5967
matr	noun_rate	0.5687
noun_pronoun_ratio	noun_rate	0.5661
noun_pronoun_ratio	function_word_rate	-0.5585

Permutation importance was computed independently within each cross-validation fold. In this way, each feature’s importance is estimated using the test fold in which it was held out.

Table 9: Permutation importance ranking of the 22 linguistic features.

Feature	Mean drop F1	Std.
function_word_rate	0.1078	0.0314
noun_rate	0.0777	0.0195
past_verb_rate	0.0632	0.0179
matrr	0.0559	0.0117
adverb_rate	0.0379	0.0108
avg_dependency_depth	0.0290	0.0052
connective_rate	0.0258	0.0071
pronoun_rate	0.0249	0.0041
adjacent_sentence_overlap	0.0206	0.0030
repetition_rate	0.0159	0.0039
subordination_rate	0.0139	0.0058
avg_word_length	0.0115	0.0039
adjective_rate	0.0064	0.0078
vague_word_rate	0.0048	0.0010
present_verb_rate	0.0034	0.0044
verb_rate	0.0029	0.0041
sentence_length_std	0.0014	0.0020
avg_sentence_length	0.0010	0.0030
noun_pronoun_ratio	0.0005	0.0020
hedge_rate	0.0004	0.0027
clause_to_sentence_ratio	0.0000	0.0014
modal_verb_rate	-0.0010	0.0002

Table 10 reports the permutation-importance values obtained independently in each fold.

Table 10: Permutation importance across the five cross-validation folds.

Feature	F1	F2	F3	F4	F5
function_word_rate	.0502	.1279	.1371	.0993	.1244
noun_rate	.0463	.0984	.0744	.0706	.0985
past_verb_rate	.0529	.0535	.0555	.0554	.0989
matrr	.0699	.0623	.0355	.0599	.0516
adverb_rate	.0170	.0397	.0465	.0458	.0407
avg_dependency_depth	.0387	.0257	.0298	.0266	.0242
connective_rate	.0337	.0201	.0189	.0353	.0209
pronoun_rate	.0252	.0204	.0206	.0268	.0313
adjacent_sentence_overlap	.0211	.0188	.0180	.0190	.0263
repetition_rate	.0216	.0155	.0108	.0186	.0129
subordination_rate	.0140	.0172	.0029	.0197	.0155

Feature	F1	F2	F3	F4	F5
avg_word_length	.0130	.0083	.0058	.0166	.0139
adjective_rate	.0198	-.0034	.0015	.0055	.0088
vague_word_rate	.0053	.0035	.0048	.0064	.0039
present_verb_rate	.0027	.0019	-.0033	.0101	.0056
verb_rate	.0106	.0000	-.0010	.0026	.0022
sentence_length_std	.0006	-.0004	-.0007	.0046	.0026
avg_sentence_length	-.0015	-.0014	.0014	.0000	.0067
noun_pronoun_ratio	-.0007	-.0014	-.0013	.0030	.0029
hedge_rate	.0003	-.0041	.0000	.0038	.0022
clause_to_sentence_ratio	.0010	-.0025	-.0004	.0008	.0012
modal_verb_rate	-.0011	-.0012	-.0010	-.0011	-.0005

6.5 Appendix E: Error Examples

Top 5 False Negatives

The following examples correspond to Alzheimer documents incorrectly classified as control with high confidence.

Table 11: Top five false negatives of the linguistic-only model.

Index	Fold	Score	Text
447	1	-3.9089	Pick Disease - Is frontotemporal Alzheimer inherited on MedicineNet
754	3	-3.3599	One missed step = 2 cracked ribs = lots of pain medication!
638	1	-3.0285	Please sign petition titled Fighting for Our Parents and Grandparents, through our Children hosted with free Petition hosting site.
809	5	-2.7564	No explanation needed.
806	1	-2.6705	came to visit me today.

The corresponding linguistic features are reported below.

Table 12: Feature values for the five false-negative examples.

Feature	447	754	638	809	806
function_word_rate	0.2500	0.1111	0.3158	0.3333	0.4000
noun_rate	0.3750	0.5556	0.3684	0.3333	0.2000
past_verb_rate	0.5000	1.0000	0.6667	1.0000	1.0000
matrr	1.0000	1.0000	0.8421	1.0000	1.0000
pronoun_rate	0.0000	0.0000	0.1053	0.0000	0.2000
avg_dependency_depth	4.0000	4.0000	7.0000	2.0000	2.0000
connective_rate	0.0000	0.0000	0.0000	0.0000	0.0000
adjacent_sentence_overlap	0.0000	0.0000	0.0000	0.0000	0.0000

High-Confidence Correct Cases

Table 13: Examples of high-confidence correct predictions.

Class	Index	Fold	Score	Text
Alzheimer	1335	5	3.5098	Friends are so great. They take your mind off of things. They laugh with you. They love you for you. I'm so thankful I have friends.
Alzheimer	155	5	3.2796	Tonight on 'Nightly News': a different kind of Alzheimer - The Daily Nightly: 'via Blog this' the media is really on top of things. God Bless, Joe
Alzheimer	275	5	3.2565	6 Advocacy: 'via Blog this' View God Bless, Joe
Control	3436	3	-4.2286	Please visit my other site- http://helpparentsagewell.com

Table 14: Feature values for the high-confidence correct examples.

Feature	1335	155	275	3436
function_word_rate	0.5385	0.3846	0.0000	0.2500
noun_rate	0.1538	0.5385	1.0000	0.0000
past_verb_rate	0.0000	0.0000	0.0000	0.0000
matr	0.7308	0.8462	1.0000	1.0000
pronoun_rate	0.3462	0.0385	0.0000	0.2500
avg_dependency_depth	2.4000	3.0000	1.0000	2.0000
connective_rate	0.4000	0.0000	0.0000	0.0000
adjacent_sentence_overlap	0.0000	0.0000	0.0000	0.0000

6.6 Appendix F: Error Analysis by Document Length

Table 15: Error rate of the linguistic-only model across document-length ranges.

Length	Documents	Alzheimer	Control	Errors	Error rate
< 20	82	67	15	25	0.3049
20-49	199	161	38	45	0.2261
50-99	284	162	122	56	0.1972
100-199	621	460	161	99	0.1594
200-499	1479	1181	298	232	0.1569
500+	979	243	736	166	0.1696

Table 16: OOF accuracy by length threshold.

Minimum length	Documents	Accuracy OOF
$\geq$ 20 words	3562	0.8321
$\geq$ 50 words	3363	0.8356
$\geq$ 100 words	3079	0.8386
$\geq$ 200 words	2458	0.8381

6.7 Appendix G: Robustness to Minimum Document Length

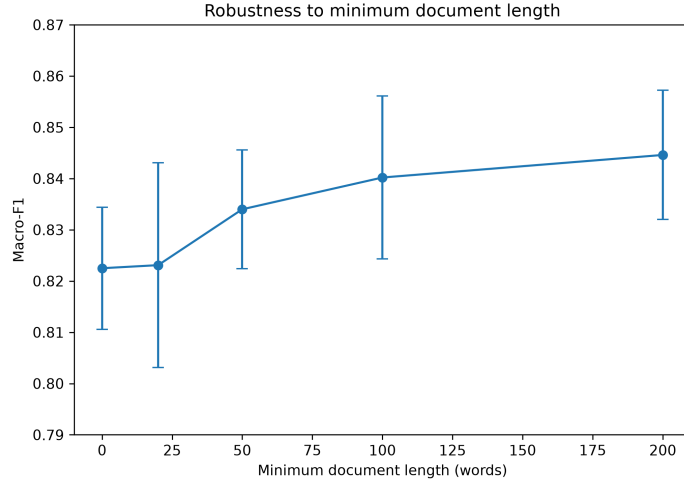


Figure 1: Length Robustness of the linguistic only model

Table 17: Performances after progressively excluding shorter documents.

Min. words	N	Dem.	Ctrl.	Macro-F1	Std.	Bal. Acc.
0	3644	2274	1370	0.8225	0.0119	0.8325
20	3562	2207	1355	0.8231	0.0200	0.8322
50	3363	2046	1317	0.8340	0.0116	0.8407
100	3079	1884	1195	0.8402	0.0159	0.8474
200	2458	1424	1034	0.8446	0.0126	0.8490

6.8 Appendix H: Compare Model Error

Table 18: Errors across the three representations.

Pattern	Documents
All correct	2956
TF-IDF + Delex only	499
TF-IDF only	81
Delex + Linguistic only	18
Delex only	8
Linguistic only	5
TF-IDF + Linguistic only	42
All wrong	35

Some examples of documents classified incorrectly by all three models are reported below:

- Do’s and Don’ts of Self-Care for New Caregivers
- Five Important Facts Veterans Need to Know about Alzheimer
- Hi Joe, ... Thank You Caregiver Nation contest ...
- Vitas HealthCare ... Advance Directives for Alzheimer’s Patients

A further interesting pattern concerns cases in which TF-IDF makes an incorrect prediction, while both models that place greater emphasis on linguistic form classify the document correctly. Some examples are reported below:

- “Well here I am again finally... I start and the old brain just stops...”
- “Well this seems to be my first initial post... I keep changing my mind or I forget what I was going to post...”

6.9 Appendix I: Dataset Summary

Table 19: Summary statistics of the dataset.

Statistic	Value
Usable documents	3644
Alzheimer-class documents	2274
Control-class documents	1370
Alzheimer mean length	277.54
Control mean length	517.37
Alzheimer median length	246.00
Control median length	526.50